Hanfei Ding

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Education

Harbin Engineering University — **Sep. 2021 – Jun. 2025**
Bachelor of Engineering in Computer Science and Technology — *Harbin, China*

Relevant Coursework

- Data Structures (95)
- Algorithm Design and Analysis (93)
- Operating Systems (91)
- Computer Organization (96)
- Artificial intelligence (95)
- Discrete Mathematics (92)
- Probability and Statistics (100)
- Mathematical Analysis (98)
- Linear Algebra (97)

Research Experience

Masked Face Recognition with Machine Learning and OpenCV — **Jul. 2023**
*Computer Vision Summer Research Program, **National University of Singapore*** — *Singapore*

- Developed a machine learning and OpenCV-based face recognition system for both masked and unmasked faces, addressing the performance degradation of conventional face recognition methods under facial occlusion.
- Constructed a masked-face dataset using facial landmarks extracted with dlib, and processed facial features with Gaussian blur, mean filtering, and SIFT-based feature extraction.
- Trained and compared SVM, KNN, and Random Forest classifiers, and designed a final decision model that achieved a 99% stranger interception rate, a 92% registered-user pass rate, and 86% overall recognition accuracy.

Deep Intelligence Laboratory, Harbin Engineering University — **Sep. 2023 – Jun. 2025**
Undergraduate Research Member; Supervisor: Prof. Rongsheng Li — *Harbin, China*

- Received systematic undergraduate research training in deep learning through technical reading, research-oriented learning, academic discussion, and project exploration under the supervision of Prof. Rongsheng Li.

Projects

Fault Classification Tool for Microservice Systems | *Python, PyTorch, GNN, Multimodal Learning* — **Jun. 2025**

- Developed a multimodal fault classification tool for microservice systems as an undergraduate thesis project, integrating logs, traces, and metrics for fault diagnosis under complex runtime environments.
- Designed feature fusion and graph representation learning modules using dilated causal convolution, multi-head attention, GLU, GAT, and GGNN to model service-level anomalies and dependencies.
- Built a parameter-shared joint classifier for anomaly detection and root cause localization, achieving the best overall performance among several baselines on the Eadro dataset and receiving the **Outstanding Undergraduate Thesis Award**.

Forum Discussion and Admin Management System | *Flask, MySQL, Redis, HTML, CSS, JavaScript* — **Nov. 2023**

- Developed a Flask-based full-stack forum system supporting user registration, login, post publishing, commenting, topic-based discussion boards, and keyword-based fuzzy search.
- Implemented the frontend with HTML, CSS, and JavaScript, and built the backend with Flask, MySQL, and Redis, including an administrator system for managing users and forum content.

Awards & Honors

Competitions

- First Prize, Mathematical Modeling League of Three Northeastern Provinces — Jul. 2024
- Second Prize, Lanqiao Cup Software Competition, Provincial Division — May 2024
- Second Prize, Mathematical Modeling League of Three Northeastern Provinces — Jul. 2023
- Third Prize, National English Competition for College Students, Category C — Oct. 2022
- First Prize, Contemporary Undergraduate Mathematical Contest in Modeling, Heilongjiang Division — Sep. 2022

Scholarships & Honors

- Outstanding Student Scholarship, Harbin Engineering University — Multiple Years
- Merit Student, Harbin Engineering University — 2021 – 2022

Technical Skills

Programming Languages: C/C++, Python, Java
Frameworks & Tools: PyTorch, Git, Docker, Maven, VS Code, Linux
Languages: English academic reading and writing